Unit: 6 Unit 7: Factoring
Topic: Finding Common Factors

Name: ____________
Date: __________

Advanced (Habanero)

Q1. What is the greatest common factor of 12 and 18?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12

Q2. Factor out the greatest common factor from $8x + 12y$
(A) $2(4x + 6y)$
(B) $4(2x + 3y)$
(C) $4x + 6y$
(D) $2x + 3y$
(E) $4(2x + 3y) + 2$

Q3. Simplify the expression $9x^2 + 12x$ by factoring
(A) $3x(3x + 4)$
(B) $3x(3x + 2)$
(C) $3x^2(3 + 4)$
(D) $3(3x^2 + 4x)$
(E) $3x(3x - 4)$

Q4. What is the greatest common factor of 24 and 30?
(A) 2
(B) 3
(C) 6
(D) 12
(E) 1

Q5. Factor out the greatest common factor from $20x^2 + 16x$
(A) $4(5x^2 + 4x)$
(B) $4x(5x + 4)$
(C) $4(5x^2 - 4x)$
(D) $4(4x^2 + 5x)$
(E) $4x^2(5 + 4)$

Q6. Simplify the expression $15x^2y + 25xy$ by factoring
(A) $5xy(3x + 5)$
(B) $5xy(3x - 5)$
(C) $5(3x^2y - 5xy)$
(D) $5xy(3x + 4)$
(E) $5(3x^2y + 5xy)$

Q7. What is the greatest common factor of 48 and 18?
(A) 2
(B) 3
(C) 6
(D) 12
(E) 1

Q8. Factor out the greatest common factor from $10x + 15y$
(A) $5(2x + 2y)$
(B) $5(2x + 3y)$
(C) $5(2x - 3y)$
(D) $2(5x + 5y)$
(E) $5(2x - 3y) + 1$

Open Ended Questions (FRQ)

Q9. Simplify the expression $12x^3 + 16x^2$ by factoring and find the value of x if the expression equals 0. Provide your answer in the following space, showing all steps.

Q10. Factor the expression $24x^4y + 36x^3y^2 + 60x^2y^3$ completely. Provide your answer in the following space, showing all steps.

Chef's Tips

- Make sure students have access to a pencil and paper
- Remind students to check their work
- Encourage students to ask for help if they need it